

CS & IT ENGINEERING



Operating System

Multithreading

One Shot

By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

Questions on Scheduling



Topics to be Covered



Topic

Multithreading

Topic

System Call: Fork()

[MCQ]

#Q. The arrival time, priority and duration of the CPU and I/O bursts for each of three processes P1, P2 and P3 are given in the table below. Each process has a CPU burst followed by an I.O burst followed by another CPU burst. Assume that each process has its own I/O resource.

Process	AT	Priority	BT (CPU)	BT(I/O)	BT(CPU)
P1	0	2	1	5	3
P2	2	3(lowest)	3	3	1
P3	3	1(highest)	2	3	1

The multi – programmed operating system uses preemptive priority scheduling. What are the finish times of the process P1, P2 and P3 ? **GATE - 2006**

A

11, 15, 9

C

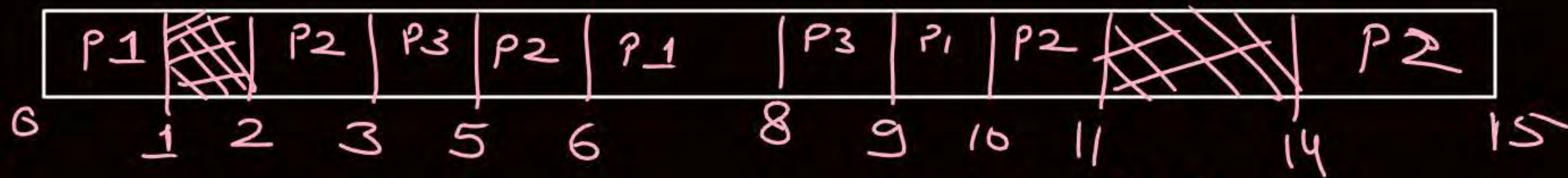
11, 16, 10

B

10, 15, 9

D

12, 17, 11



$$\begin{array}{|l} \text{P1 I/O} \\ \hline 1+5 = 6 \end{array} \rightarrow$$

$$\begin{array}{|l} \text{P3 I/O} \\ \hline 5+3 = 8 \end{array} \rightarrow$$

$$\text{CPU idle} = \frac{4}{15} * 100\%$$

$$\text{CPU used} = \frac{11}{15} * 100\%$$

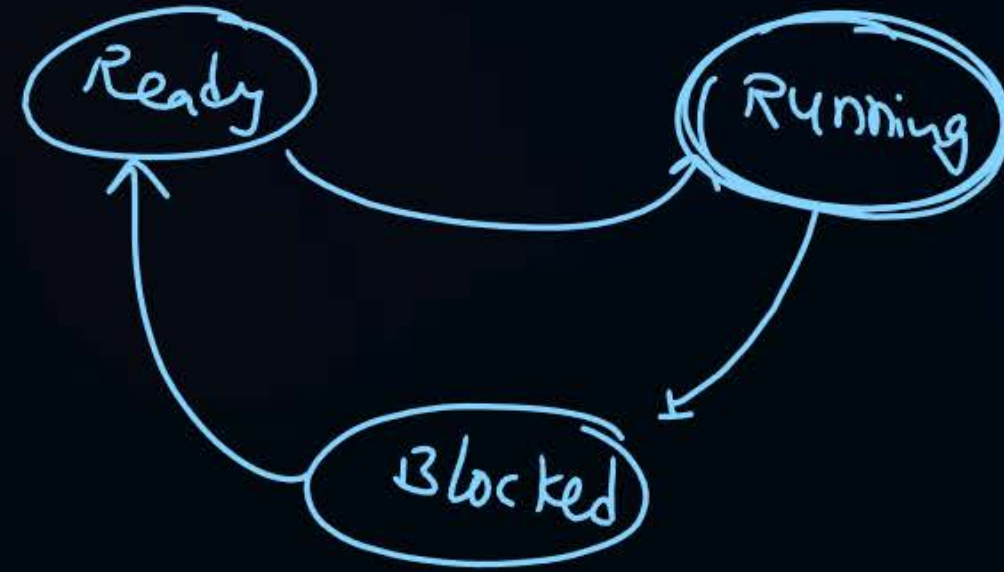


Topic : CPU Utilization



1. Without IO Operations

$$\text{CPU utilization}^n = 1$$



2. With IO Operations

assume a process wants to use I/O for p fraction of time.

$$n=1 \Rightarrow \text{CPU utilization} = 1 - p$$

$$n=2 \Rightarrow = 1 - p^2 \leftarrow \text{when both processes go for I/O}$$

$$n=3 \Rightarrow = 1 - p^3$$

$$n \vdots \boxed{= 1 - p^n}$$

#Q. A computer system has degree of multiprogramming 10. And maximum I/O wait, that can be tolerated is 0.5. The CPU utilization is %?

↓
99.9

$$= 1 - (0.5)^{10}$$

$$= 0.999$$

$$= 99.9\%$$

#Q. Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst Time	3	1	3	$Z = 2$

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is _____.

Gate - 2019

$$\text{avg } w_T = \frac{\text{Total } w_T \text{ of all process}}{4}$$

$$1 \text{ ms} = \frac{\text{Total } w_T}{4} \Rightarrow \text{Total } w_T = 4 \text{ ms}$$

P1	P2	P1	P4	P3	
0	1	2	4	6	9

	BT
P3	3
P4	2

	WT
P1	1
P2	0
P3	$1 + \textcircled{2} = 2$
P4	

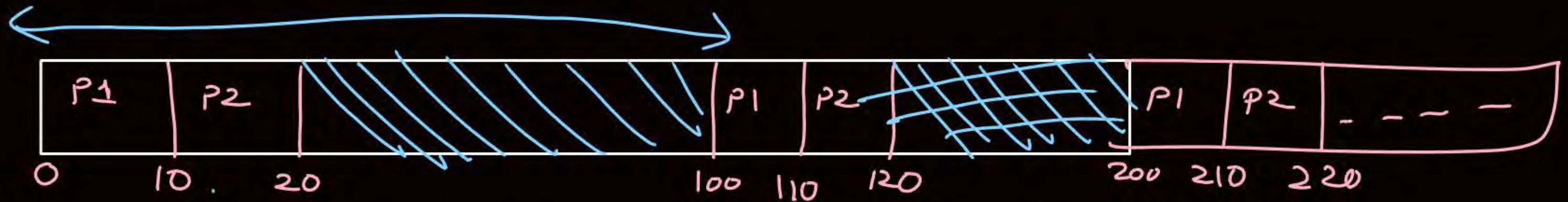
} 4

[MCQ]

#Q. A uni-processor computer system only has two processes, both of which alternate 10 ms CPU bursts with 90 ms I/O bursts. Both the process were created at nearly the same time. The I/O of both processor can proceed in parallel. Which of the following scheduling strategies will result in the least CPU utilization (over a long period of time) for this system ?

GATE - 2003

- A** First come first served scheduling
- B** Shortest remaining time first scheduling
- C** Static priority scheduling with different priorities for the two processes
- D** ✓ Round robin scheduling with a time quantum of 5 ms.



$$10 + 90 = 100$$

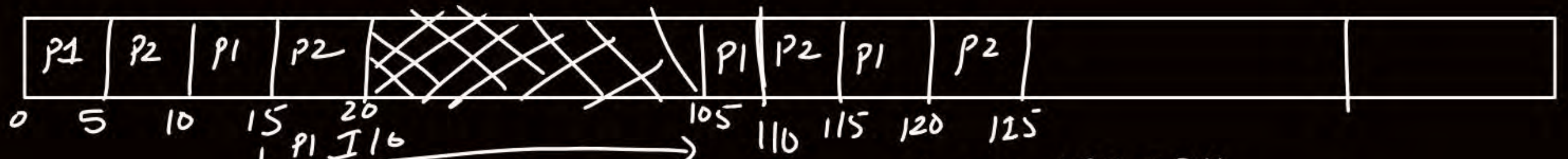
$$P2 \quad 20 + 90 = 110$$

$$110 + 90 = 200$$

$$120 + 90 = 210$$

$$\text{CPU utilization} = \frac{20}{100} = 0.2 \text{ or } 20\%$$

Round Robin:-



$$\text{CPU utilization} = \frac{20}{105}$$

$$15 + 90 = 105$$

$$P2 \quad 20 + 90 = 110$$

$$120 + 90 = 210$$

$$125 + 90 = 215$$

#Q. Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

GATE - 2006

0%



10.6%



30.0%



89.4%

#Q. Consider four processes P, Q, R, and S scheduled on a CPU as per round-robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

GATE-2022

- (A) $P = 4, Q = 10, R = 6, S = 2$
- (B) $P = 2, Q = 9, R = 5, S = 1$
- (C) $P = 4, Q = 12, R = 5, S = 4$
- (D) $P = 3, Q = 7, R = 7, S = 3$



Topic : Multithreading



Component of process

or

Lightweight Process

Provide a way to improve application performance through parallelism

sum = 0;

for (i = 0; i < 100000; i++)

{

sum = sum + i;

}

⇓

single CPU,
single process

⇓
A lot of time needed



4 CPU

single process

⇓

A lot of time needed
(one process runs on
one CPU)

4 CPU

4 processes

first 25K next 25K next 25K next 25K
P1 P2 P3 P4

⇓
parallel work

→ now final sum calculated
→ 4 processes will take lot of space

multithreading

↓
make 4 threads
of single process

→ very less time

T₁ ⇒ first 25K sum

T₂ ⇒ next 25K sum

T₃ ⇒ — 11 — 11 —

T₄ ⇒ last 25K sum

→ process
does
the final
sum

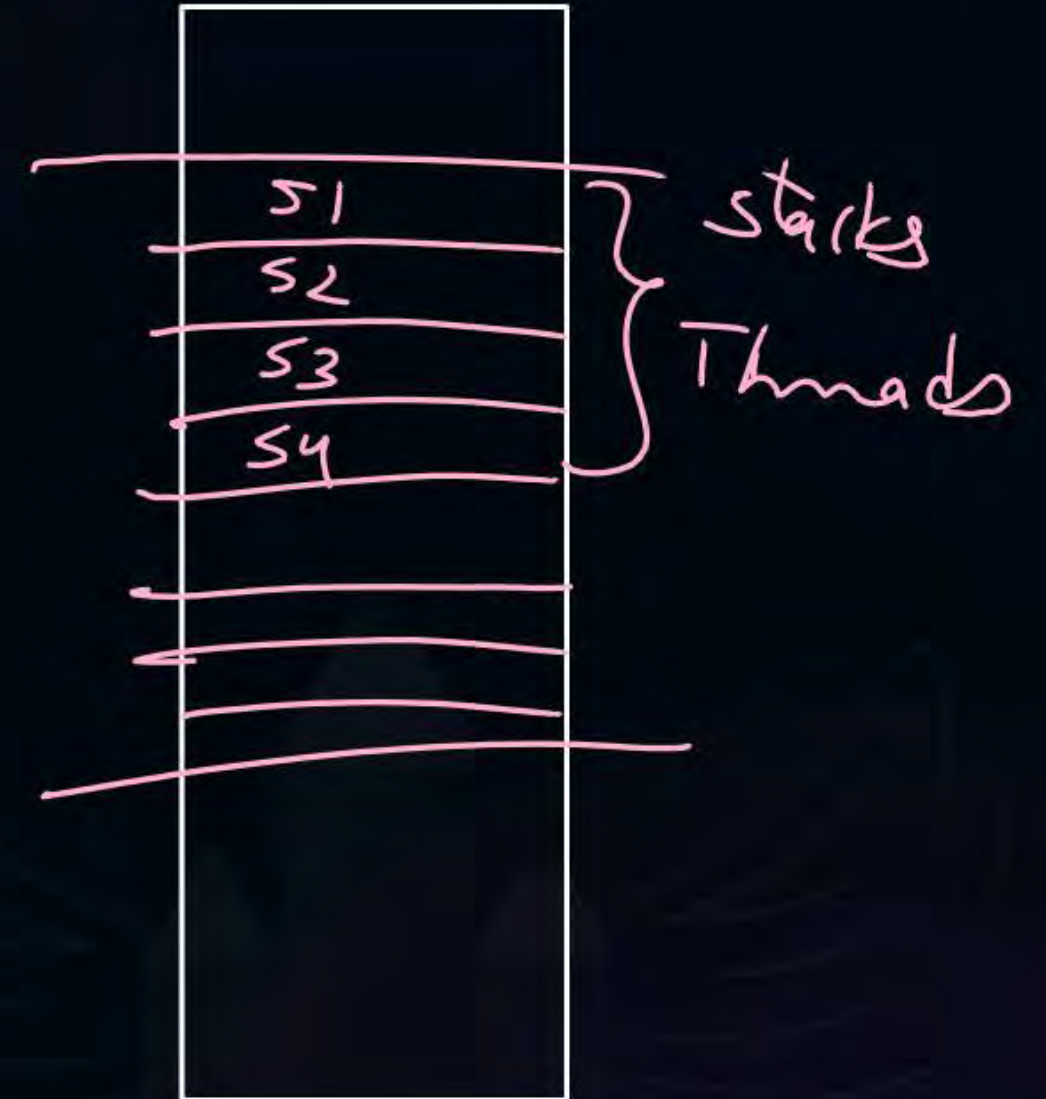
→ all 4 threads can run on 4 CPUs parallelly.



Topic : Threads

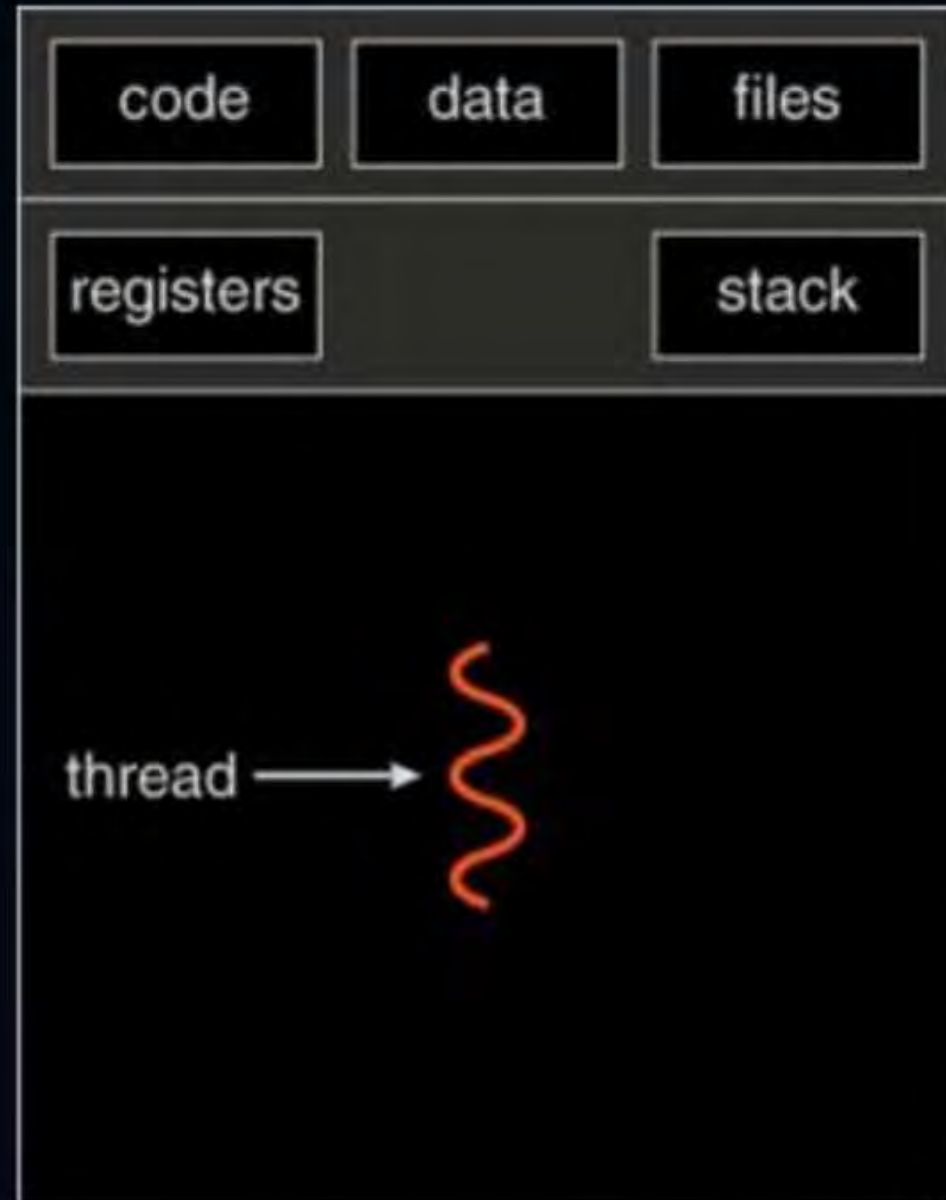


Shared Among Threads	Unique For Each Thread
Code Section	Thread Id
Data Section	Register Set
OS Resources	Stack
Open Files & Signals	Program Counter
Heap	

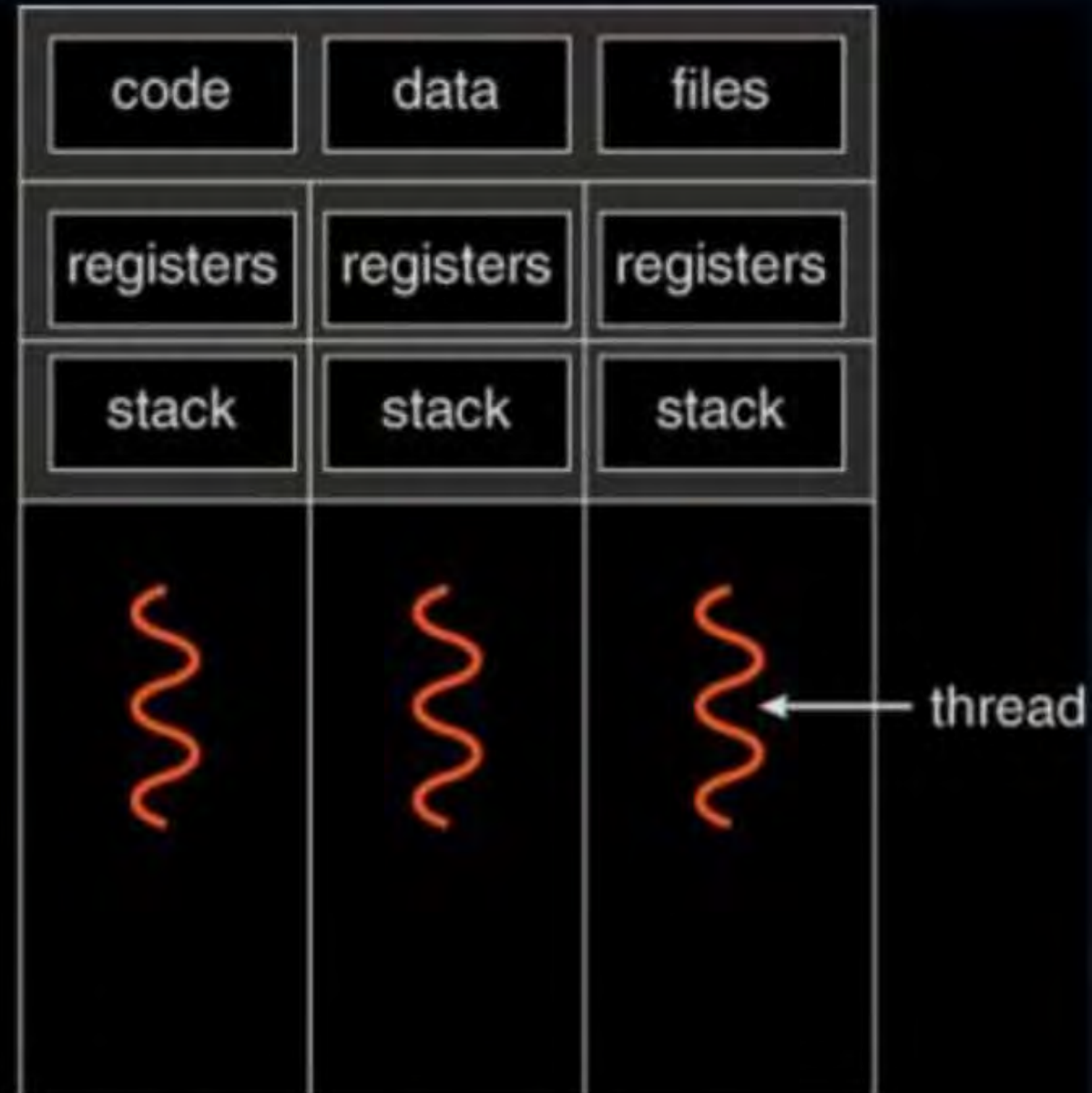




Topic : Threads



single-threaded process



multithreaded process



Topic : Advantages of Multithreading



- Responsiveness
- Faster Context Switch
- Resource Sharing
- Economy
- Communication
- Utilization of Multiprocessor Architecture





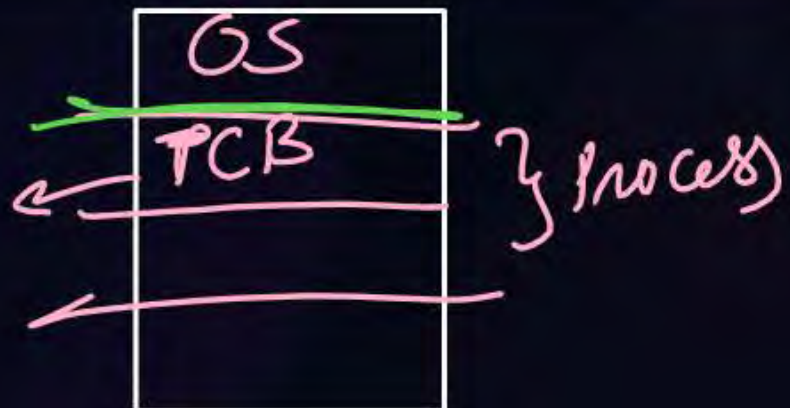
Topic : Types of Threads

1. User Level Thread
2. Kernel Level Thread



Topic : Types of Threads

User Threads	Kernel Thread
Multithreading in user process	Multithreading in kernel process
Created without kernel intervention	Kernel itself is multithreaded
Context switch is very fast	Context switch is slow
If one thread is blocked, OS blocks entire process	Individual thread can be blocked
Generic and can run on any OS	Specific to OS
Faster to create and manage	Slower to create and manage



TCB $\Rightarrow$ Thread Control block



Topic : System Call



Programmatic way in which a computer program requests a service from kernel





Topic : How System Call Works



System Call

	Windows	Unix
Process Control	CreateProcess() ExitProcess() WaitForSingleObject()	fork() exit() wait()
File Manipulation	CreateFile() ReadFile() WriteFile() CloseHandle()	open() read() write() close()
Device Manipulation	SetConsoleMode() ReadConsole() WriteConsole()	ioctl() read() write()
Information Maintenance	GetCurrentProcessID() SetTimer() Sleep()	getpid() alarm() sleep()
Communication	CreatePipe() CreateFileMapping() MapViewOfFile()	pipe() shmget() mmap()
Protection	SetFileSecurity() InitializeSecurityDescriptor() SetSecurityDescriptorGroup()	chmod() umask() chown()



Topic : Fork() System Call

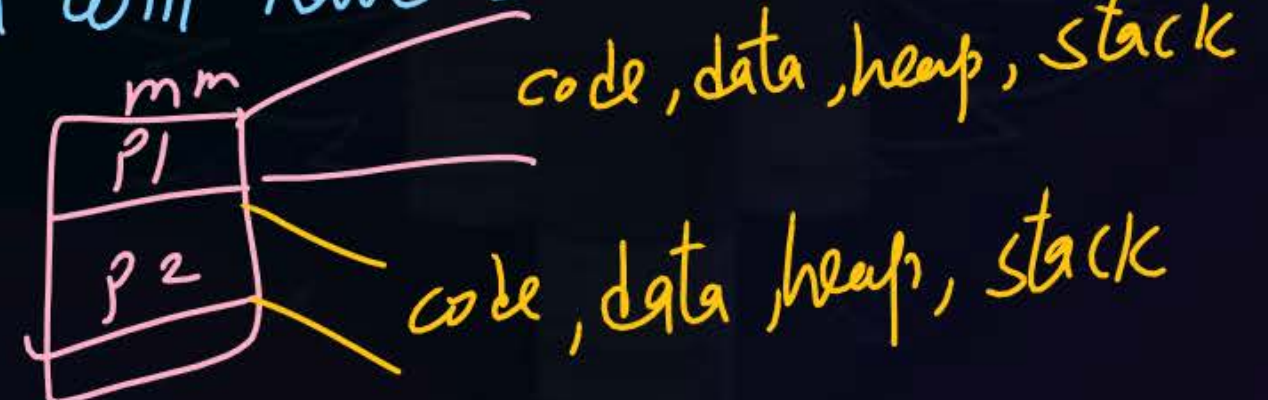
Fork system call is used for creating a new process, which is called child process.

Child process runs concurrently with the process that makes the fork() call (parent process).

$P1 \Rightarrow \text{fork}() \Rightarrow P2$

$P2$ is child process $P1$

$P1$ and $P2$ both will have same code.



Ex:-

P1
fork()
printf("*");

child process starts executing
from next "inst" after the fork()
which created it.





Topic : Fork() System Call

It takes no parameters and returns an integer value

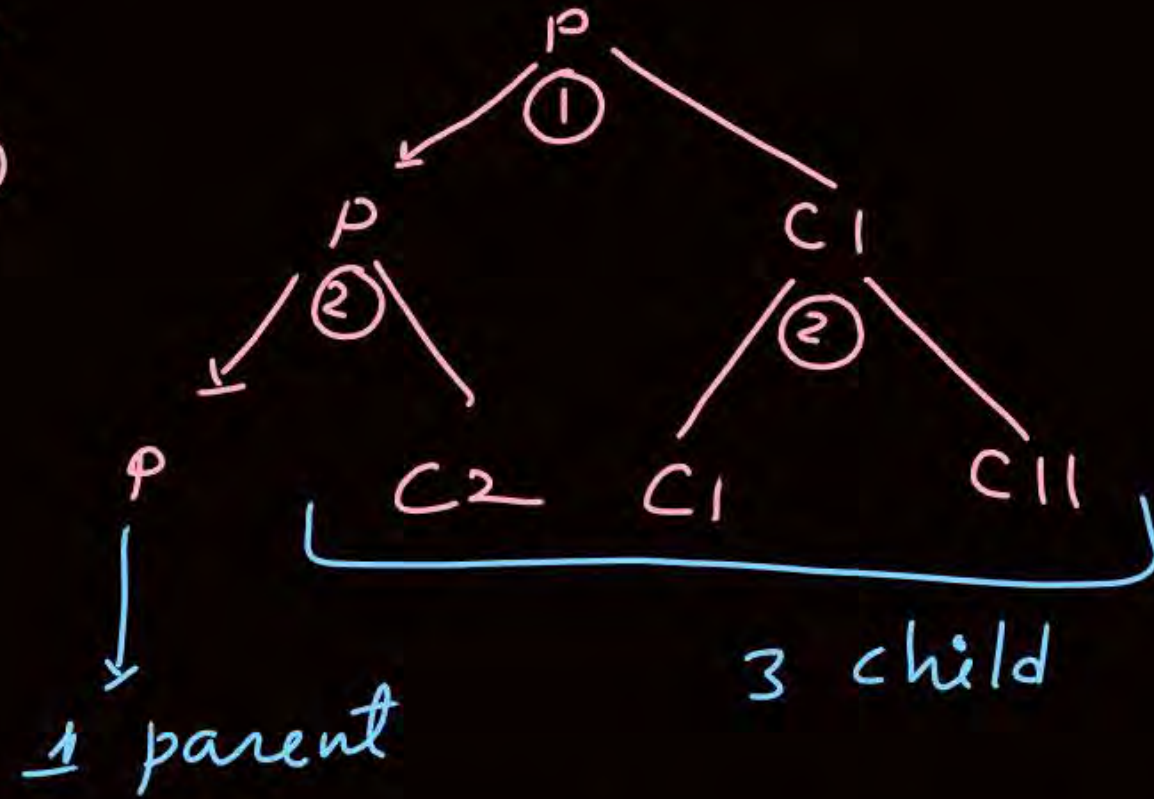
- Negative Value: (-1)
Creation of a child process was unsuccessful

- Zero:
Returned to the newly created child process
- Positive value:
Returned to parent or caller. The value contains process ID of newly created child process

→ successful

Ques)

fork(); ①
fork(); ②



Total $\Rightarrow$ 4 processes

[MCQ]



#Q. A process executes the code

```
{  
fork();  
fork();  
fork();  
}
```

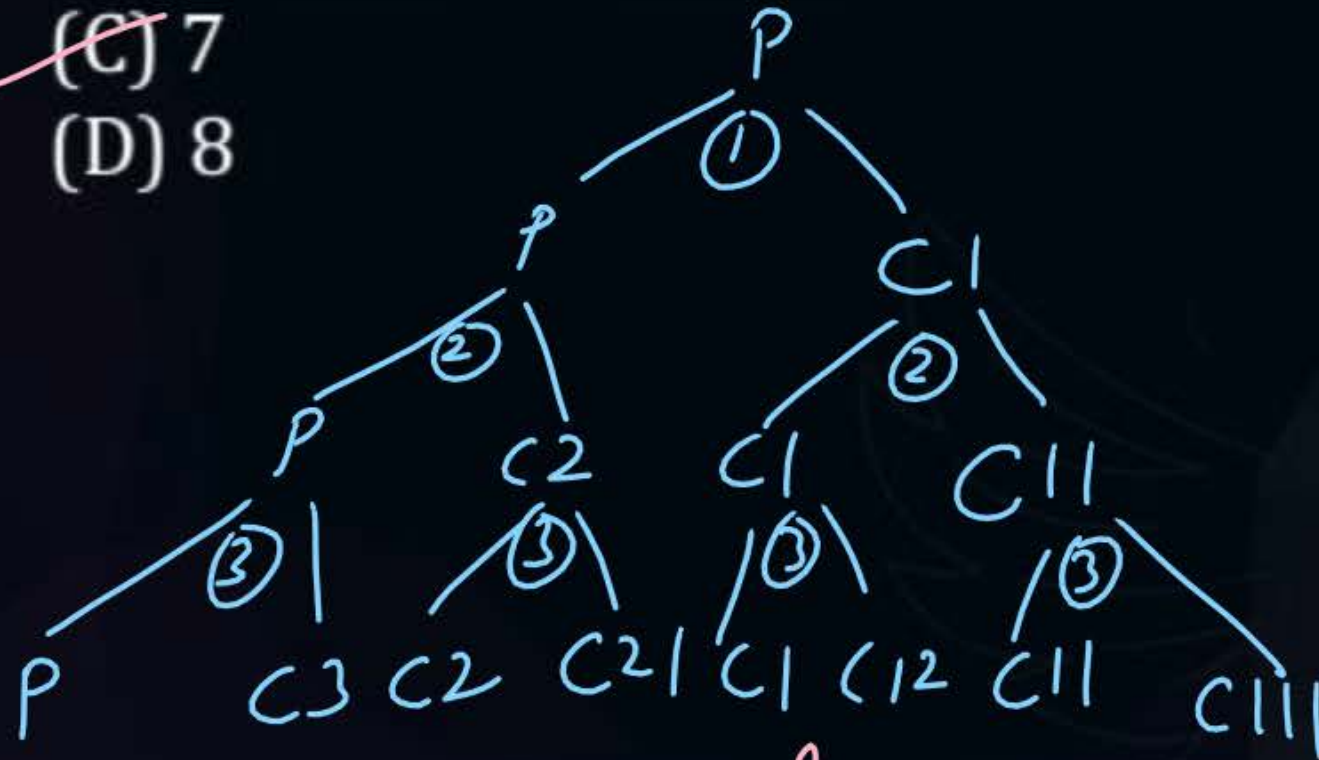
The total number of child processes created is?

(A) 3

(B) 4

☒ (C) 7
(D) 8

GATE-2008



1 parent
7 child processes

[NAT]

#Q. A process executes the code
`fork ();`
`:`
`fork ();`

There are n such statements. The total number of child processes created is?

$$\text{Total child processes} = 2^n - 1$$

$$\text{Total processes including parent} = 2^n$$



2 mins Summary

Topic

Multithreading

Topic

System Call: Fork()



Happy Learning

THANK - YOU

